Linear equations in two variables — Part 1

Algebra · Math · 65 questions

1.1 103a78aa-4206-460e-8583-9fb2c2c88e63 · Easy

$$y = 70x + 8$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	8
2	148
4	288

B.

x	y
0	70
2	78
4	86

C.

x	y
0	70
2	140
4	280

D.

x	y
0	8
2	132

4	272

1.2 84a52145-ed23-4de2-9bb6-c1f412e7b1f8 · Easy

A line in the *xy*-plane has a slope of $-\frac{1}{2}$ and passes through the point $0, 3$. Which equation represents this line?

A. $y = -\frac{1}{2}x - 3$

B. $y = -\frac{1}{2}x + 3$

C. $y = \frac{1}{2}x - 3$

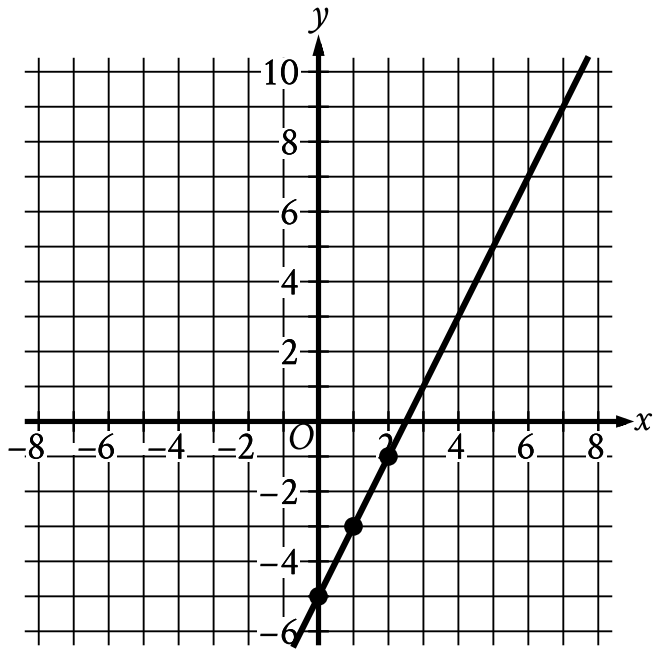
D. $y = \frac{1}{2}x + 3$

1.3 43d2ddd5-a11f-494c-acca-88dc80fde548 · Easy

Line *k* is defined by $y = \frac{1}{4}x + 1$. Line *j* is parallel to line *k* in the *xy*-plane. What is the slope of *j*?

Student-produced response: _____

1.4 243f2e8b-7dd9-4a6f-b01e-582ee9fbb34c · Easy



- The line slants sharply up from left to right.
- The line passes through the following points:
  - (0 comma negative 5)
  - (1 comma negative 3)
  - (2 comma negative 1)

The graph shows the linear relationship between x and y. Which table gives three values of x and their corresponding values of y for this relationship?

A.

x	y
0	0
1	-7
2	-9

B.

x	y
0	0
1	-3
2	-1

C.

x	y
0	-5
1	-7
2	-9

D.

x	y
0	-5
1	-3
2	-1

1.5 fb1fc3f7-9487-4a08-841a-d7c028989341 · Easy

A chemist studying the impact of salt on a process mixes x kilograms of a low-salt mixture, which is 2% salt by weight, with y kilograms of a high-salt mixture, which is 96% salt by weight, to create 24 kilograms of a mixture that is 4% salt by weight. Which equation represents this situation?

A. $0.96x + 0.02y = 0.0424$

B. $0.02x + 0.96y = 0.0424$

C. $0.96x + 0.02y = 24$

D. $0.02x + 0.96y = 24$

1.6 e3d6b3f4-d0ec-46bc-b1d7-45b9bfe60aae · Easy

$$8x + 11y = 170$$

The equation gives the possible combinations of the number of 2009 premium grade Log Cabin Pennies, x, and the number of 1996 select grade Lincoln Pennies, y, in a collection that is worth a total of $\$170$. If there are 6 1996 select grade Lincoln Pennies in the collection, how many 2009 premium grade Log Cabin Pennies are in the collection?

Student-produced response: _____

1.7 451c1b50-026a-4bad-96b2-c090526934bf · Easy

Line k is defined by $y = 3x + 15$. Line j is perpendicular to line k in the xy-plane. What is the slope of line j?

A. $-\frac{1}{3}$

B. $-\frac{1}{12}$

C. $-\frac{1}{18}$

D. $-\frac{1}{45}$

1.8 c2ed48a0-96b6-4990-b199-6a704301200c · Easy

The equation $46 = 2x + 2y$ gives the perimeter of a rectangular rug that has length x, in feet, and width y, in feet. The width of the rug is 8 feet. What is the length, in feet, of the rug?

Student-produced response: _____

1.9 b0517abf-3415-4fde-bf1b-cc79e78acc97 · Easy

Line r in the xy-plane has a slope of 4 and passes through the point $0, 6$. Which equation defines line r?

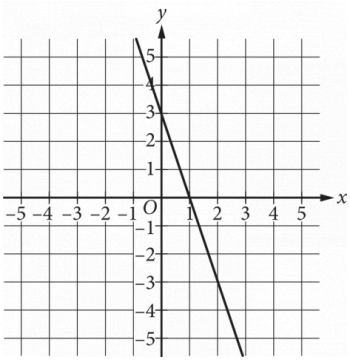
A. $y = -6x + 4$

B. $y = 6x + 4$

C. $y = 4x - 6$

D. $y = 4x + 6$

1.10 09915-DC · Easy



What is the equation of the line shown in the xy-plane above?

A. $y = 3x - 3$

B. $y = -3x + 3$

C. $y = \frac{1}{3}x - 3$

D. $y = -\frac{1}{3}x + 3$

1.11 032760-DC · Easy

A city's total expense budget for one year was x million dollars. The city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between x and y in this context?

A. $x+y=201$

B. $x-y=201$

C. $2x-y=201$

D. $y-x=201$

1.12 62aec139-478e-41ed-ae31-7c2ce572a8ef · Easy

The y-intercept of the graph of $y = -6x - 32$ in the xy-plane is $0, y$. What is the value of y?

Student-produced response: _____

1.13 023151-DC · Easy

The equation $y=0.1x$ models the relationship between the number of different pieces of music a certain pianist practices, y, during an x-minute practice session. How many pieces did the pianist practice if the session lasted 30 minutes?

A. 1

B. 3

C. 10

D. 30

1.14 bae4968b-c35c-406d-b991-c09a53f7410e · Easy

In 2010, a swim club had a total of 35 swimmers, each classified as either advanced or intermediate. From 2010 to 2020, the number of advanced swimmers in the club increased by approximately 53%, and the number of intermediate swimmers in the club increased by approximately 44%. The total number of swimmers in the club increased by approximately 49%. Which equation best represents this situation, where a represents the number of advanced swimmers in the club in 2010 and b represents the number of intermediate swimmers in the club in 2010?

A. $1.53a + 1.49b = 35(1.44)$

B. $1.49a + 0.53b = 35(1.44)$

C. $1.53a + 1.44b = 35(1.49)$

D. $1.44a + 1.53b = 35(1.49)$

1.15 dcbc1809-77bc-40b1-b74a-cb105d2b0efa · Easy

A line in the xy-plane has a slope of $\frac{1}{9}$ and passes through the point $0, 14$. Which equation represents this line?

A. $y = -\frac{1}{9}x - 14$

B. $y = -\frac{1}{9}x + 14$

C. $y = \frac{1}{9}x - 14$

D. $y = \frac{1}{9}x + 14$

1.16 56ef2f00-3bce-4eb9-bc8b-077ee70a054d · Easy

The equation $46 = 2a + 2b$ gives the relationship between the side lengths a and b of a certain parallelogram. If $a = 9$, what is the value of b?

Student-produced response: _____

1.17 b5bc3c68-cf0b-420c-894c-67a87a23e1b4 · Easy

A teacher is creating an assignment worth 70 points. The assignment will consist of questions worth 1 point and questions worth 3 points. Which equation represents this situation, where x represents the number of 1-point questions and y represents the number of 3-point questions?

A. $4xy = 70$

B. $4x + y = 70$

C. $3x + y = 70$

D. $x + 3y = 70$

1.18 7a8622f4-3e4b-4373-a170-e62a46579a04 · Easy

The equation $40x + 20y = 160$ represents the number of sweaters, x, and number of shirts, y, that Yesenia purchased for $\$ 160$. If Yesenia purchased 2 sweaters, how many shirts did she purchase?

A. 3

B. 4

C. 8

D. 40

1.19 2c79adc0-8360-4810-80d7-655a0eb1e8f9 · Easy

A producer is creating a video with a length of 70 minutes. The video will consist of segments that are 1 minute long and segments that are 3 minutes long. Which equation represents this situation, where x represents the number of 1-minute segments and y represents the number of 3-minute segments?

A. $4xy = 70$

B. $4x + y = 70$

C. $3x + y = 70$

D. $x + 3y = 70$

1.20 affdca44-b865-4102-a2b3-d54b3337bd47 · Easy

$$y = x + 4$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	4
1	5
2	6

B.

x	y
0	6
1	5
2	4

C.

x	y
0	2
1	1
2	0

D.

x	y
0	0
1	1

2	2

1.21 9040ee10-4498-4a27-b92f-5181148b3e11 · Easy

Davio bought some potatoes and celery. The potatoes cost $\$0.69$ per pound, and the celery cost $\$0.99$ per pound. If Davio spent $\$5.34$ in total and bought twice as many pounds of celery as pounds of potatoes, how many pounds of celery did Davio buy?

A. 2

B. 2.5

C. 2.67

D. 4

1.22 62a434cb-ba25-40cd-9fae-dadf2a410594 · Easy

Last week, an interior designer earned a total of $\$1{,}258$ from consulting for x hours and drawing up plans for y hours. The equation $68x + 85y = 1{,}258$ represents this situation. Which of the following is the best interpretation of 68 in this context?

A. The interior designer earned $\$68$ per hour consulting last week.

B. The interior designer worked 68 hours drawing up plans last week.

C. The interior designer earned $\$68$ per hour drawing up plans last week.

D. The interior designer worked 68 hours consulting last week.

1.23 2c084f8b-ca91-4384-99bb-5c4f22a12de2 · Easy

In a chess tournament, each participant earns 1 point for each game the participant plays that ends in a draw and 3 points for each game the participant wins. A certain participant in this tournament has earned 41 points. Which equation represents this situation, where d represents the number of games this participant has played that ended in a draw and w represents the number of games this participant has won?

A. $d + 3w = 41$

B. $3d + w = 41$

C. $d + \frac{w}{3} = 41$

D. $\frac{d}{3} + w = 41$

1.24 025936-DC · Easy

In the xy-plane, a line has a slope of 6 and passes through the point $(0,8)$. Which of the following is an equation of this line?

A. $y = 6x + 8$

B. $y = 6x + 48$

C. $y = 8x + 6$

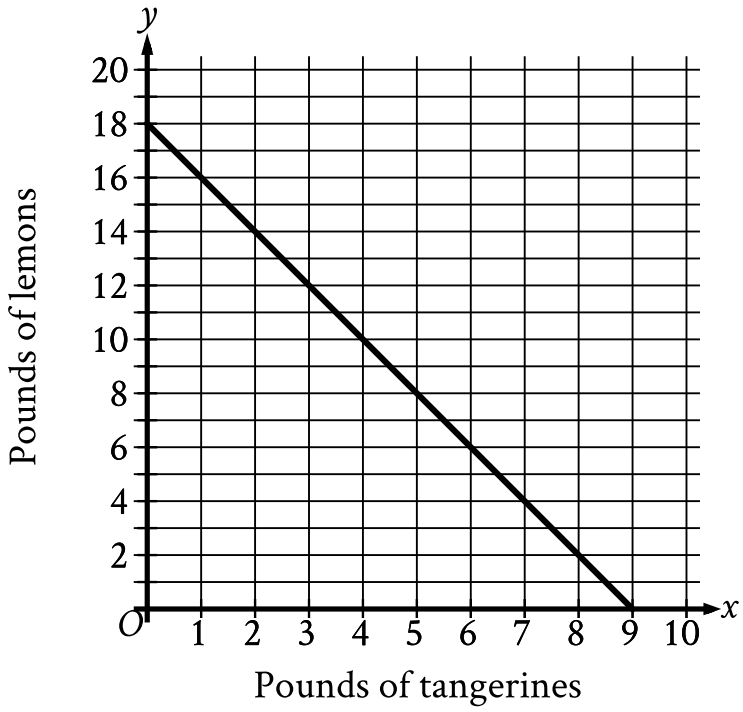
D. $y = 8x + 48$

1.25 7e0f2970-3e0b-4e25-b69c-84a25543e4e7 · Easy

Vivian bought party hats and cupcakes for $\$ 71$. Each package of party hats cost $\$ 3$, and each cupcake cost $\$ 1$. If Vivian bought 10 packages of party hats, how many cupcakes did she buy?

Student-produced response: _____

1.26 1f20ceb9-1f8d-417b-b1f8-a18ac789333a · Easy



- The line slants sharply down from left to right.
- The line passes through the following points:
  - (0 comma 18)
  - (4 comma 10)
  - (7 comma 4)
  - (9 comma 0)

The graph shows the possible combinations of the number of pounds of tangerines and lemons that could be purchased for $\$ 18$ at a certain store. If Melvin purchased lemons and 4 pounds of tangerines for a total of $\$ 18$, how many pounds of lemons did he purchase?

A. 7

B. 10

C. 14

D. 16

1.27 eee8837f-1da3-4242-8e86-e8d7f893b7c4 · Easy

Naomi bought both rabbit snails and nerite snails for a total of $\$ 52$. Each rabbit snail costs $\$ 8$ and each nerite snail costs $\$ 6$. If Naomi bought 2 nerite snails, how many rabbit snails did she buy?

A. 5

B. 12

C. 14

D. 50

1.28 7375a642-6151-451c-8a97-8c5092f8e9b5 · Easy

Jay walks at a speed of 3 miles per hour and runs at a speed of 5 miles per hour. He walks for *w* hours and runs for *r* hours for a combined total of 14 miles. Which equation represents this situation?

A. $3w + 5r = 14$

B. $\frac{1}{3}w + \frac{1}{5}r = 14$

C. $\frac{1}{3}w + \frac{1}{5}r = 112$

D. $3w + 5r = 112$

1.29 05414-DC · Easy

A gardener buys two kinds of fertilizer. Fertilizer A contains 60% filler materials by weight and Fertilizer B contains 40% filler materials by weight. Together, the fertilizers bought by the gardener contain a total of 240 pounds of filler materials. Which equation models this relationship, where x is the number of pounds of Fertilizer A and y is the number of pounds of Fertilizer B?

A. $0.4x + 0.6y = 240$

B. $0.6x + 0.4y = 240$

C. $40x + 60y = 240$

D. $60x + 40y = 240$

1.30 5b3a2276-e0f1-47e6-a1cd-e590bbf22925 · Easy

The equation $x + y = 1{,}440$ represents the number of minutes of daylight (between sunrise and sunset), x, and the number of minutes of non-daylight, y, on a particular day in Oak Park, Illinois. If this day has 670 minutes of daylight, how many minutes of non-daylight does it have?

A. 670

B. 770

C. $1{,}373$

D. $1{,}440$

1.31 cc275a34-d420-4dc2-8b26-a526a2d95c32 · Easy

What is the equation of the line that passes through the point $0, 5$ and is parallel to the graph of $y = 7x + 4$ in the xy-plane?

A. $y = 5x$

B. $y = 7x + 5$

C. $y = 7x$

D. $y = 5x + 7$

1.32 e144d7c1-e5c3-4695-b317-58369ec6c22e · Easy

$$x + y = 350$$

The given equation relates the total number of maple trees, x, and the total number of birch trees, y, planted in a 14-acre forest preserve. If 245 maple trees were planted in the forest preserve, how many birch trees were planted in the forest preserve?

A. 14

B. 25

C. 105

D. 245

1.33 08464-DC · Easy

$x + y = 75$

The equation above relates the number of minutes, x, Maria spends running each day and the number of minutes, y, she spends biking each day. In the equation, what does the number 75 represent?

A. The number of minutes spent running each day

B. The number of minutes spent biking each day

C. The total number of minutes spent running and biking each day

D. The number of minutes spent biking for each minute spent running

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1.34 06912-DC · Easy

x	y
1	5
2	7
3	9
4	11

The table above shows some pairs of x values and y values. Which of the following equations could represent the relationship between x and y ?

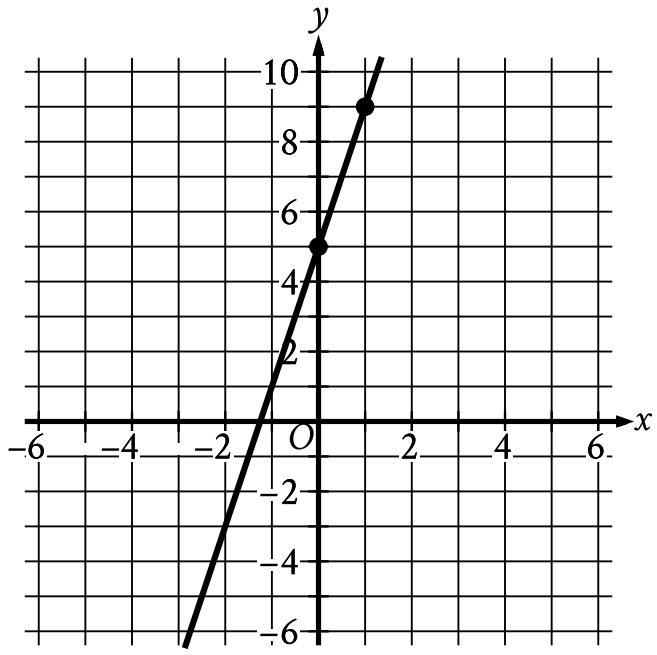
A. $y = 2x + 3$

B. $y = 3x - 2$

C. $y = 4x - 1$

D. $y = 5x$

1.35 d57fb5b1-3f4b-4846-927d-9d44e8d5aa05 · Easy



- The line slants sharply up from left to right.
- The line passes through the following points:
  - (0 comma 5)
  - (1 comma 9)

Line j is shown in the xy-plane. Line k (not shown) is parallel to line j. What is the slope of line k?

Student-produced response: _____

1.36 025553-DC · Easy

$3a+4b=25$

A shipping company charged a customer $25 to ship some small boxes and some large boxes. The equation above represents the relationship between a, the number of small boxes, and b, the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

A. 3

B. 4

C. 5

D. 6

1.37 db170278-9037-413e-b5e2-ed81bbf21899 · Easy

A mixture consisting of only vitamin D and calcium has a total mass of 150 grams. The mass of vitamin D in the mixture is 50 grams. What is the mass, in grams, of calcium in the mixture?

A. 200

B. 150

C. 100

D. 50

1.38 7d5c4716-5975-4df0-abca-00aaa9789ed0 · Easy

An employee at a restaurant prepares sandwiches and salads. It takes the employee 1.5 minutes to prepare a sandwich and 1.9 minutes to prepare a salad. The employee spends a total of 46.1 minutes preparing x sandwiches and y salads. Which equation represents this situation?

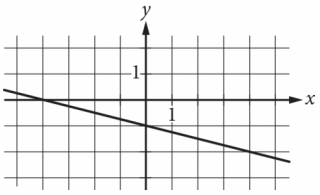
A. $1.9x + 1.5y = 46.1$

B. $1.5x + 1.9y = 46.1$

C. $x + y = 46.1$

D. $30.7x + 24.3y = 46.1$

1.39 016625-DC · Easy



Which of the following is an equation of the graph shown in the xy-plane above?

A. $y = -\frac{1}{4}x - 1$

B. $y = -x - 4$

C. $y = -x - \frac{1}{4}$

D. $y = -4x - 1$

1.40 025710-DC · Easy

$F = 2.50x + 7.00y$

In the equation above, F represents the total amount of money, in dollars, a food truck charges for x drinks and y salads. The price, in dollars, of each drink is the same, and the price, in dollars, of each salad is the same. Which of the following is the best interpretation for the number 7.00 in this context?

A. The price, in dollars, of one drink

B. The price, in dollars, of one salad

C. The number of drinks bought during the day

D. The number of salads bought during the day

1.41 54986bed-7fc7-4f46-a3c6-d49d4fe54080 · Easy

x	y
0	18
1	13
2	8

The table shows three values of x and their corresponding values of y. There is a linear relationship between x and y. Which of the following equations represents this relationship?

A. $y = 18x + 13$

B. $y = 18x + 18$

C. $y = -5x + 13$

D. $y = -5x + 18$

1.42 aaebe46b-7143-4bb9-ae10-3f9ce2a0fbcb · Easy

A total of 364 paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. What is the best interpretation of $x, y = 24, 73$ in this context?

A. If 24 triangles were constructed, then 73 rectangles were constructed.

B. If 24 triangles were constructed, then 73 paper straws were used.

C. If 73 triangles were constructed, then 24 rectangles were constructed.

D. If 73 triangles were constructed, then 24 paper straws were used.

1.43 e9060623-fc35-41b6-8e81-702b67a6dc3c · Easy

For a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water, for a total of 240 liters of water. Which equation represents this situation?

A. $x + 3y = 240$

B. $x + y = 240$

C. $3x + 3y = 240$

D. $3x + y = 240$

1.44 61a4ffe2-81eb-435b-bda6-4f44122ffded · Easy

A shipment consists of 5-pound boxes and 10-pound boxes with a total weight of 220 pounds. There are 13 10-pound boxes in the shipment. How many 5-pound boxes are in the shipment?

A. 5

B. 10

C. 13

D. 18

1.45 992a511e-c690-479d-bab1-f58211e97302 · Easy

$$7x - 4y = -84$$

For the given equation, which table gives three values of x and their corresponding values of y?

A.

x	0	4	8
y	21	28	35

B.

x	0	4	8
y	35	28	21

C.

x	21	28	35
y	0	4	8

D.

x	21	28	35
y	8	4	0

1.46 70115054-fb11-47dc-8abe-e8208d6f27f6 · Easy

A food truck buys forks for $\$ 0.04$ each and plates for $\$ 0.48$ each. The total cost of x forks and y plates is $\$ 661.76$. Which equation represents this situation?

A. $0.48x - 0.04y = 661.76$

B. $0.04x - 0.48y = 661.76$

C. $0.48x + 0.04y = 661.76$

D. $0.04x + 0.48y = 661.76$

1.47 63b423d6-d760-4ccc-974a-f9ad2904f1e9 · Easy

The y-intercept of the graph of $12x + 2y = 18$ in the xy-plane is $0, y$. What is the value of y?

Student-produced response: _____

1.48 023185-DC · Easy

$4x + 3y = 24$

Mario purchased 4 binders that cost x dollars each and 3 notebooks that cost y dollars each. If the given equation represents this situation, which of the following is the best interpretation of 24 in this context?

A. The total cost, in dollars, for all binders purchased

B. The total cost, in dollars, for all notebooks purchased

C. The total cost, in dollars, for all binders and notebooks purchased

D. The difference in the total cost, in dollars, between the number of binders and notebooks purchased

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1.49 694cab4d-2a4c-40a2-bd61-46ca0d6708b4 · Easy

Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the xy-plane and passes through the point $0, 5$. Which equation defines line j?

A. $y = 6x + 5$

B. $y = -5x + 5$

C. $y = -6x + 5$

D. $y = 5x + 5$

1.50 025561-DC · Easy

Tony spends $80 per month on public transportation. A 10-ride pass costs $12.50, and a single-ride pass costs $1.50. If g represents the number of 10-ride passes Tony buys in a month and t represents the number of single-ride passes Tony buys in a month, which of the following equations best represents the relationship between g and t ?

A. $g + t = 80$

B. $g + t = 1.50 + 12.50$

C. $1.50g + 12.50t = 80$

D. $12.50g + 1.50t = 80$

1.51 a63a0a09-9923-4dab-b757-04fffd0b40b1 · Easy

$$y = -4x + 40$$

Which table gives three values of x and their corresponding values of y for the given equation?

A.

x	y
0	0
1	-4
2	-8

B.

x	y
0	40
1	44
2	48

C.

x	y
0	40
1	36
2	32

D.

x	y
0	0
1	4

2	8

1.52 afbd028c-6ff7-478a-83f7-9d7942bf56c6 · Easy

A machine makes large boxes or small boxes, one at a time, for a total of 700 minutes each day. It takes the machine 10 minutes to make a large box or 5 minutes to make a small box. Which equation represents the possible number of large boxes, x, and small boxes, y, the machine can make each day?

A. $5x + 10y = 700$

B. $10x + 5y = 700$

C. $x + y10 + 5 = 700$

D. $10 + x5 + y = 700$

1.53 012f9eba-ac05-4ad4-b0df-834b624098e7 · Easy

What is the y-intercept of the graph of $y = 34x + 81$ in the xy-plane?

A. $0, 81$

B. $0, 34$

C. $0, -34$

D. $0, -81$

1.54 0b8822c0-8322-4969-8780-d3d067076d46 · Easy

A store sells two different-sized containers of a certain Greek yogurt. The store's sales of this Greek yogurt totaled $1{,}277.94$ dollars last month. The equation $5.48x + 7.30y = 1{,}277.94$ represents this situation, where x is the number of smaller containers sold and y is the number of larger containers sold. According to the equation, which of the following represents the price, in dollars, of each smaller container?

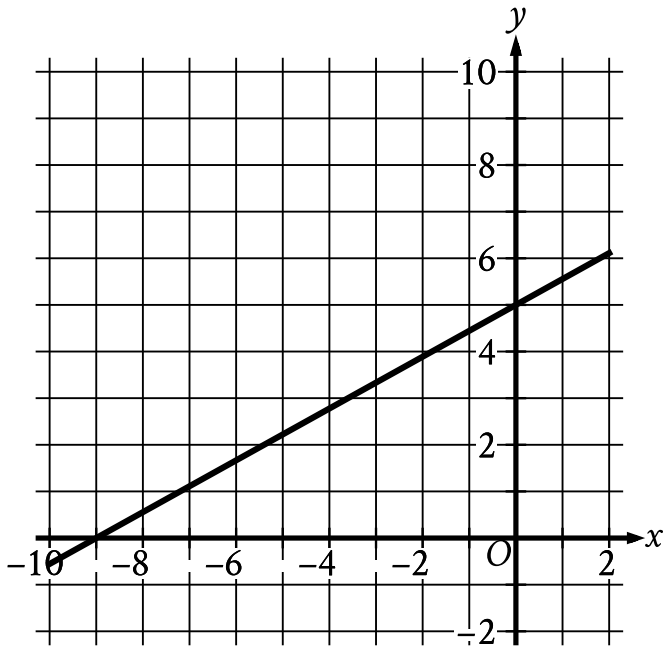
A. 5.48

B. $7.30y$

C. 7.30

D. $5.48x$

1.55 d8fb921c-f434-41b5-b963-d87e3d7e5311 · Easy



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (negative 9 comma 0)
  - (0 comma 5)

What is the y-intercept of the line graphed?

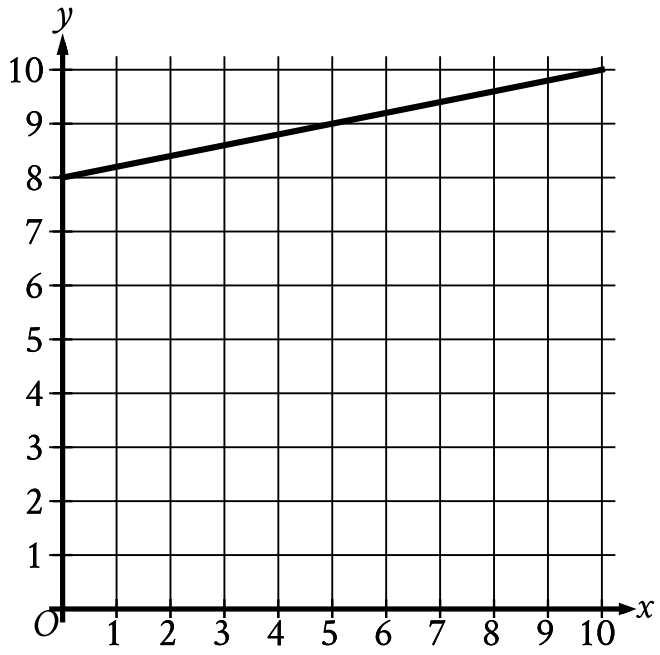
A. $-5, 0$

B. $0, 0$

C. $0, 5$

D. $0, 9$

1.56 6c6e9e8c-6e47-4600-94eb-347c7a7ae6c5 · Easy



- The line slants gradually up from left to right.
- The line passes through the following points:
  - (0 comma 8)
  - (5 comma 9)
  - (10 comma 10)

What is the y-intercept of the line graphed?

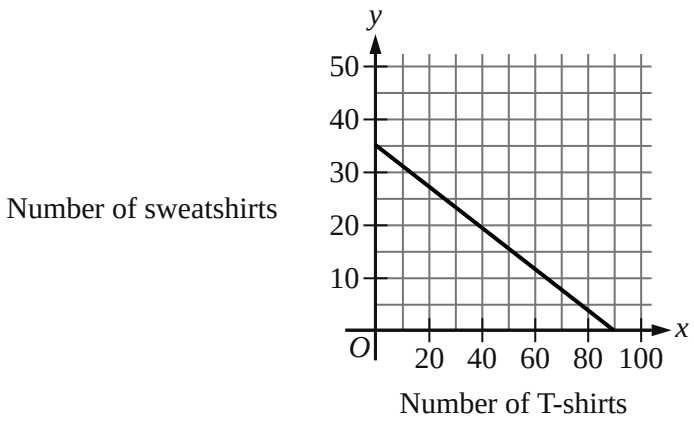
A. $0, -8$

B. $0, -\frac{1}{8}$

C. $0, 0$

D. $0, 8$

1.57 6bfc998d-ffc0-408c-a5a9-fdf16efed66f · Medium



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (0 comma 35)
  - (90 comma 0)

The graph models the relationship between the number of T-shirts, x, and the number of sweatshirts, y, that Kira can purchase for a school fundraiser. Which equation could represent this relationship?

A. $y = 7x + 18$

B. $7x + 18y = 630$

C. $y = 18x + 7$

D. $18x + 7y = 630$

1.58 cd071339-36d3-4977-8ff2-863f8066f9ed · Medium

A batch of banana milkshakes consists of 4 cups of ice cream and 2 bananas and has $1{,}114$ milligrams mg of calcium. There is 276 mg of calcium in 1 cup of the ice cream used to make this batch of milkshakes. How much calcium, in mg, is in 1 banana?

A. 5

B. 10

C. 419

D. $1{,}104$

1.59 65ac5dc5-af25-4ecf-8562-a346047750a6 · Medium

Line k is defined by $y = -\frac{17}{3}x + 5$. Line j is perpendicular to line k in the xy-plane. What is the slope of line j?

Student-produced response: _____

1.60 ea6362be-d602-4fd3-b6df-53518625197a · Medium

Figure A and figure B are both regular polygons. The sum of the perimeter of figure A and the perimeter of figure B is 63 inches. The equation $3x + 6y = 63$ represents this situation, where x is the number of sides of figure A and y is the number of sides of figure B. Which statement is the best interpretation of 6 in this context?

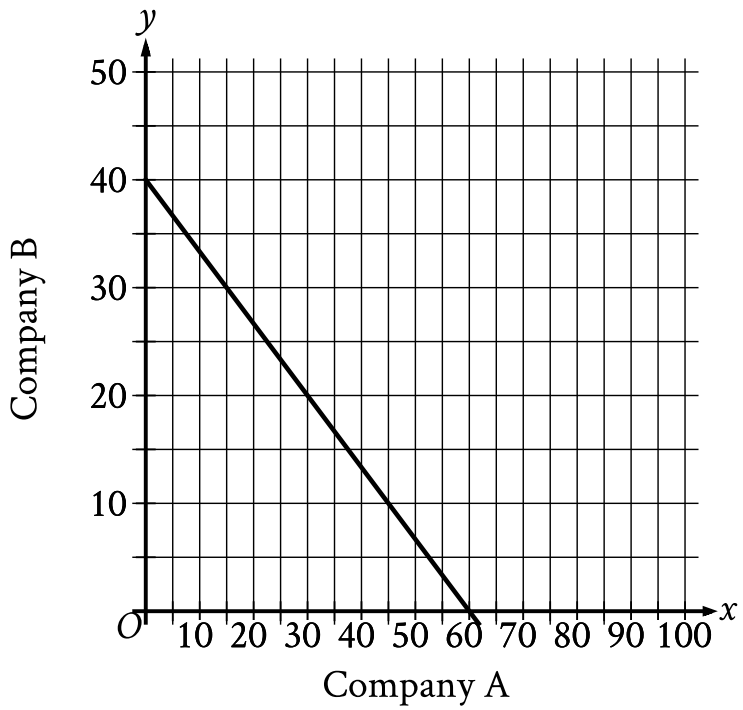
A. Each side of figure B has a length of 6 inches.

B. The number of sides of figure B is 6.

C. Each side of figure A has a length of 6 inches.

D. The number of sides of figure A is 6.

1.61 64377d96-584c-4863-b696-51d7230fc57d · Medium



- The line slants down from left to right.
- The line passes through the following points:
  - (0 comma 40)
  - (60 comma 0)

The graph shows the relationship between the number of shares of stock from Company A, x, and the number of shares of stock from Company B, y, that Simone can purchase. Which equation could represent this relationship?

A. $y = 8x + 12$

B. $8x + 12y = 480$

C. $y = 12x + 8$

D. $12x + 8y = 480$

1.62 7ebcee2e-7a57-4d43-ae10-fe0a44f7791e · Medium

A certain township consists of a 5-hectare industrial park and a 24-hectare neighborhood. The total number of trees in the township is $4{,}529$. The equation $5x + 24y = 4{,}529$ represents this situation. Which of the following is the best interpretation of x in this context?

A. The average number of trees per hectare in the industrial park

B. The average number of trees per hectare in the neighborhood

C. The total number of trees in the industrial park

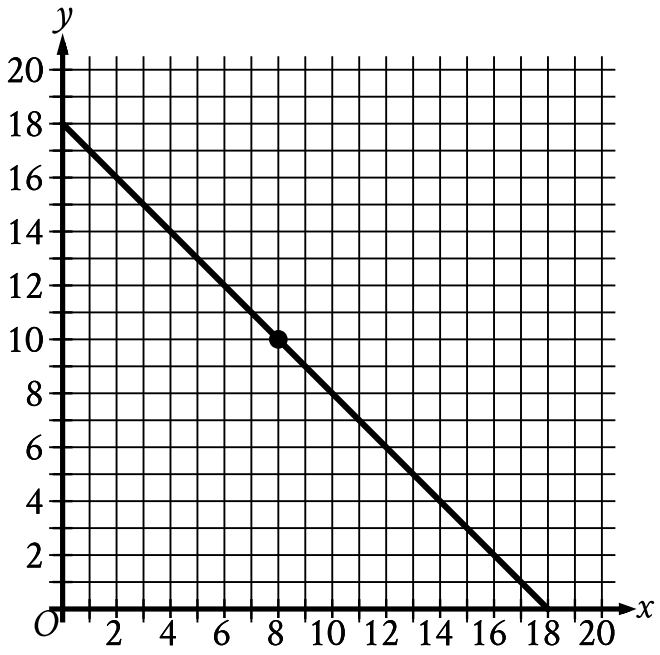
D. The total number of trees in the neighborhood

1.63 d4f91dd3-90d3-4caa-a259-64740bd3f7e3 · Medium

In the xy-plane, line p has a slope of $-\frac{5}{3}$ and an x-intercept of $(-6, 0)$. What is the y-coordinate of the y-intercept of line p?

Student-produced response: _____

1.64 4d7cd4c5-d45b-40fb-8391-76021972ba43 · Medium



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (8 comma 10)
  - (9 comma 9)

The graph in the *xy*-plane models the possible combinations of length x, in meters (m), and width y, in meters, for a rectangle with a perimeter of 36 m. Which statement is the best interpretation of the point $8, 10$ in this context?

A. The length is 10 m less than the perimeter, and the width is 8 m less than the perimeter.

B. The length is 10 m, and the width is 8 m.

C. The length is 8 m, and the width is 10 m.

D. The length is 8 m less than the perimeter, and the width is 10 m less than the perimeter.

1.65 029899-DC · Medium

On a 210-mile trip, Cameron drove at an average speed of 60 miles per hour for the first x hours. He then completed the trip, driving at an average speed of 50 miles per hour for the remaining y hours. If $x = 1$, what is the value of y ?

Student-produced response: _____